Programme Structure

Faculty	Science and Technology	**Pattern**	*2024*
Department	Computer Science	**Date (w.e.f.)**	*06/01/2025*
Programme	B.Sc. – Computer Science		

Semester -II									
Course Type	**Course Code**	**Course Name**	**Teaching Scheme (Hours/Week)**			**Credit**	**Examination Scheme and Marks**		
			L	**T**	**P**	**C**	**CIE**	**ESE**	**Total**
DSC	BSCCS24201	Introduction to Python Programming	2	0	0	2	30	20	50
DSC	BSCCS24202	Introduction to Python Programming Lab	0	0	2	1	15	10	25
DSC	BSCCS24203	Database Management System	2	0	0	2	30	20	50
DSC	BSCCS24204	Database Management System Lab	0	0	2	1	15	10	25
CC	BSCCS24205	Capstone Project-II	0	0	4	2	50	0	50
MC	MT24CS02A	VU Level - Web Technologies	2	0	2	3	60	40	100
OE	OE24CS02A	VU Level - Applications of IoT	2	1	0	3	60	40	100
SEC	SE24CS02A	VU Level - Digital Literacy	1	1	0	2	50	0	50
IKS	IKS24CS01A	VU Level - Transformation & Applications of Information Technology	2	0	0	2	50	0	50
VEC	VE24CS02A	VU Level- Environmental Studies	2	1	2	4	60	40	100
TOTAL			**13**	**3**	**12**	**22**	**420**	**180**	**600**
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

BSCCS24203 : Database Management System

Course Type	DSC	Semester	2

Teaching Scheme		Credits		Examination Scheme	
Lecture	2Hrs./Week	Lecture	2	CIE Marks	60
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	40
Practical/Studio	2 Hrs./Week	Practical/Studio	1	Total Marks	100
Total	4 Hrs./Week	Total	3		

Course Objectives

CO No.	At the end of this course students will be able to …..
1	To study the concept of data ,database and database management system
2	To study DML,DDL,TCL and DCL languages for data based operation
3	To learn the concept of database design using E-R diagram and relational algebra
4	To study normalization technique, transaction processing, concurrency Control
5	To study concepts of PL/SQL programming

Course Outcomes

CO No.	Statement
1	Understand fundamental concepts of DBMS.
2	Design E-R Diagram for any system
3	Understand various normalization technique and relational algebra
4	Perform various operation on database using DML and DDL commands
5	Develop constructs of PL/SQL programming for Database applications

Mapping of COs to POs and PSOs

CO Number	POs												PSOs			BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	
CO1	1	2	1										1			3
CO2	1	1											1			2
CO3	2												1	1		3
CO4	2	2											1			2
CO5	2	2											1			2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom’s Taxonomy Level

COURSE CONTENTS

UNIT-1	**Introduction to DBMS**	**6 hrs**	**CO**	**BTL**
Data, Information, Database and introduction to DBMS, RDBMS Concept, Characteristics and Advantages of RDBMS, Architecture of DBMS, Three views of data mode, Physical, Conceptual and external model, Data independence, Database users, Database administrator, Data dictionary.			1	1,2 1
UNIT-2	**E-R Diagram**	**6 hrs**		
Entity Relationship Model, Entity Types, Entity Sets, Attributes and its types, Keys, E-R Diagram, Key constraints, mapping constraints, Unary, Binary and ternary relationship, Aggregation and generalization, Data Integrity			2	2, 3,4
UNIT-3	**Database Design and Relational Algebra**	**6 hrs**		
Database Design , Functional dependencies, Normalization (1NF, 2NF, 3 NF and BCNF), Integrity constraints, Relational Algebra operations selection, projection, union, intersection, Cartesian product, types of joins			3	2,3,4 ,5,6
UNIT-4	**SQL**	**6 hrs**		
Introduction, DDL, DML, TCL and DCL commands, Specifying constraints in SQL, Aggregate functions, Built-in functions – numeric, date, string functions, Sub-queries, Use of group by, having, order by			4	3, 4
UNIT-5	**Transaction Management, Concurrency Control and PL/SQL programming**	**6 hrs**		
Introduction to Transaction Processing, Desirable Properties of Transactions, Concepts of Concurrency Control, PL/SQL programming, procedures, functions, cursor, triggers. Case Study on Practical Learning with HackerRank for Databases to strengthen practical understanding of Database Management Systems (DBMS) and to develop problem-solving skills, gain real-world experience in writing and optimizing queries.			5	5

Course Content

List of Experiments/ Studio Assignments, etc

Sr. No	**Name of Experiment/ Studio Assignment**	**CO**	**BTL**
1	Case Study on Practical SQL Learning Using HackerRank hands-on experience with Structured Query Language (SQL) by solving real-world SQL challenges, to improve query-writing skills, understand database optimization, and gain practical knowledge essential for industry applications. https://www.hackerrank.com/domains/sql	1, 2, 3, 4	1, 3
2	Lab Assignment 1: MySQL – Introduction, Installing MySQL, Verifying MySQL Installation, Post-installation Steps, Setting Up a MySQL User Account, Administrative MySQLCommand	1,2	1, 3

3	Lab Assignment 2 : Create Database, Select your database, USE database, Display a list of your availabledatabases Data Definition Language commands, Data Manipulation Language commands- Basic queries Design and Develop SQL DDL & DML statements to demonstrate create, alter, drop,delete, select, insert, update commands for given	2,3	

	database schema.		
4	Lab Assignment 3: Data Definition Language commands And Data Manipulation Language commands- Basic queries Create Book, publisher, Author tables. 1. Insert at least five records in each table. 2. Add column address in the Book_authors table. 3. Display particular record from table using where clause. 4. Display all books whose prize is greater than RS 100/- from BOOK table. 5. Delete particular record from BOOK_AUTHOR table. 6. Update particular record from BOOK table. 7. Add column in BOOK_AUTHOR using alter command.	2, 3	2
5	Lab Assignment 4: Create tables salesman having schema. salesman(sales_id,salesman_name,salesman_address,sales_done) customer(customer_id,customer_name,customer_address)) Establish 1 to M relationship between two tables and insert atleast 5 records and display 5 records in table execute following queries 1) Find the avg salesdone 2) Find the customers staying in Mumbai 3) Create index 4) Drop index created 5) Find salesmam name who has done maximum sale	2,4	4
6	Lab Assignment 5 : Create table empoyee having schema(EId, Ename, Salary, Joindate) Insert 5 records and execute the following queries: 1. Write an sql statement to update the salary to 70,000, where employee name is damodar. 2. Delete from the employee data whose have joined on 09-05-2022. 3. Add the following DOB into employee table 4. Execute date functions : curdate(), CURRENT_TIMESTAMP() and DATEPART()	2, 4	5, 3
7	Lab Assignment 6: Write following PL/SL programs Print numbers 1 to 20 using exit when loop Print odd numbers between 1 to 20 using for loop Print odd numbers between 1 to 20 using while loop Write PL/SQL function max to find maximum of two numbers.	5	6

8	Lab Assignment 7: Use of group by,having order by clause and built in functions	2, 4	4

Textbooks

1	Database management systems, Rghu Ram Krishnan
2	Introduction to database management systems,C. J. Date

Reference Books / Journal Articles / Weblink

1	Data base System Concepts, Silberschatz, Korth, McGraw hill, Sixth Edition.
2	Database management systems, Tennan baum
3	http://nptel.ac.in/